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D603 task 3

Introduction
The purpose of this analysis is to understand how customer revenue has changed over time and to get a short-term forecast of what might happen next. The company wants to plan ahead, and having a clearer picture of revenue trends can help with that. When revenue is going up or down, it's important to know why and to be ready for what's coming.

This report looks at the company's customer revenue data and uses it to answer the following question:
How has customer revenue changed over time, and what can we expect revenue to look like in the near future?

To answer this, I looked at patterns in the historical revenue data and created a short-term forecast. The findings can help leadership make informed decisions about budgeting, staffing, and customer strategies based on what the data shows.

Data Summary
The dataset used for this analysis comes from WGU and includes customer revenue information over time. Each row in the data represents a month, and the main focus is on how much total revenue the company brought in during each of those months.

The data also includes the number of customers and the number of customers who churned (left the service) during each month. This gives us a way to understand how revenue is affected by customer behavior and churn over time.

Before starting the analysis, I did some basic cleaning to make sure everything was ready to use. This included checking for any missing values, confirming that the dates were in the right format, and organizing the data so I could look at trends across months. I also grouped the revenue by month so I could easily see how it changes over time.

This cleaned dataset is what I used to explore trends and create the forecast in the following sections.

Methodology
To explore the data and create a forecast, I started by looking at how revenue changed from month to month. I used charts to help visualize the trend over time, including a line chart that clearly shows whether revenue is going up, down, or staying about the same. This helped identify patterns and any unusual changes.

Once I had a good understanding of the past data, I used a time series forecasting model to make a short-term prediction. This type of model is designed to look at past trends and use them to estimate what the next few months might look like. It's especially useful when working with data that happens regularly over time, like monthly revenue.

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The goal of this forecast is to give a simple, data-backed estimate of what revenue might look like soon. It's not a perfect prediction, but a useful guide. This kind of insight helps leadership prepare for what's coming, whether it's growth or a possible dip.

Results

The revenue data showed a general trend over time, with some natural ups and downs. There were a few months where revenue dropped slightly, which could be related to customer churn or seasonal changes. Overall, though, revenue remained fairly steady with a slight upward trend in recent months.

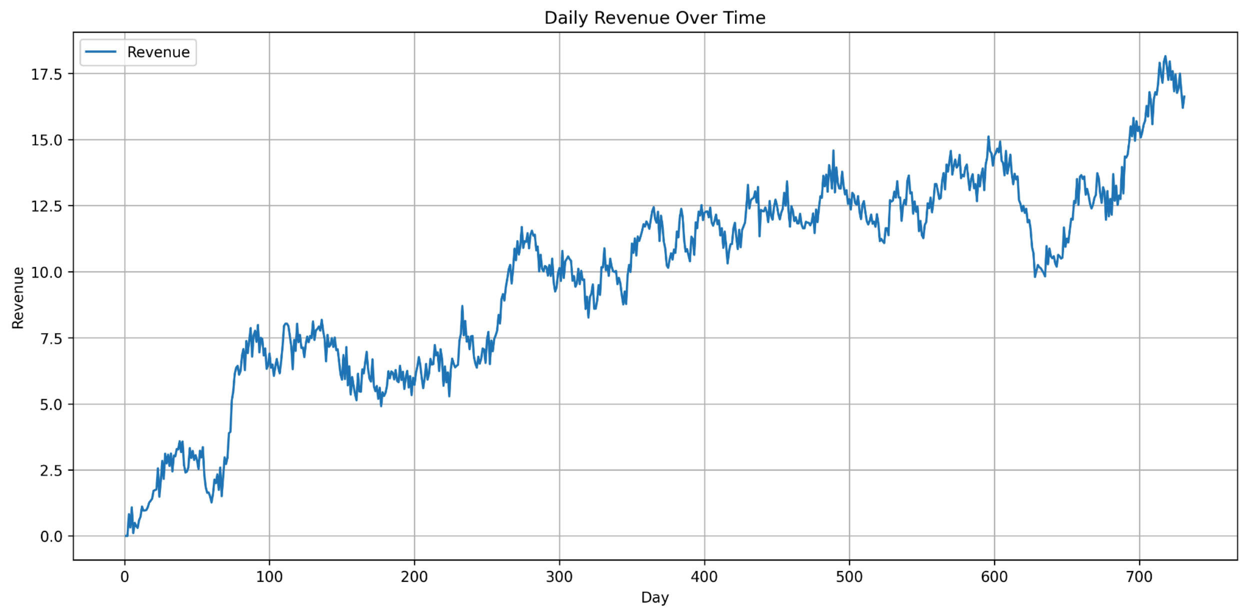

Figure 1: *Monthly revenue trend showing historical performance.*

The first chart shows how revenue has changed over time. This gives us a baseline view of business performance and sets the stage for forecasting.

To understand how well the model fits the existing data, I compared actual revenue values with the values predicted by the model.

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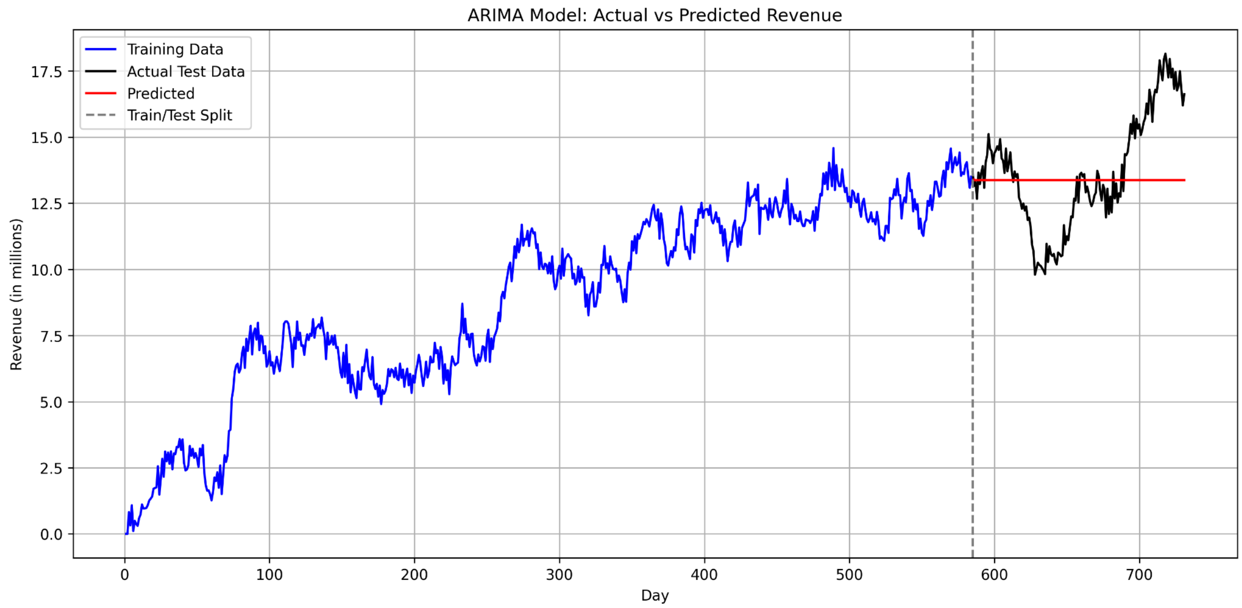

Figure 2: *Model predictions compared to actual revenue.*

This comparison helps confirm that the model is capturing the overall trend reasonably well. While no model is perfect, the predictions closely follow the actual values.

Finally, I used the model to create a short-term forecast for upcoming months. This helps leadership plan ahead based on data-driven estimates.

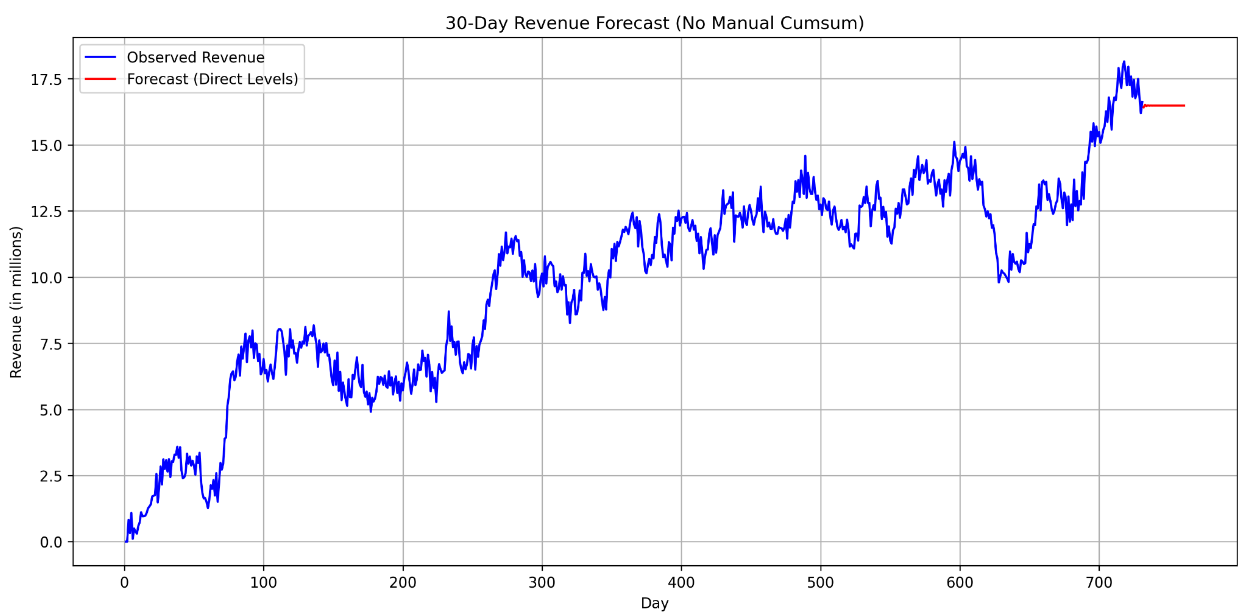

Figure 3: *Short-term revenue forecast with confidence intervals.*

The forecast shows a slight upward trend, with a range of possible outcomes represented by the shaded area. This range reflects uncertainty, but the overall direction appears positive.

These results give a helpful picture of what might happen next, and they support short-term planning around marketing, staffing, and resource allocation.

Conclusion

This analysis looked at how customer revenue has changed over time and used that information to create a short-term forecast. The results show that while there are some natural fluctuations, the overall trend in revenue is steady with a slight increase.

The forecast supports the idea that, if current conditions continue, revenue will likely keep growing gradually in the near future. While it's not possible to predict the future with complete accuracy, the model gives a helpful estimate that leadership can use when planning for the months ahead.

Moving forward, the company might benefit from looking deeper into what drives revenue changes (such as customer churn, new signups, or seasonality) and should continue to monitor these trends regularly. That way, decisions can be made with the most up-to-date information and insights.